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A Layered Agent Fund

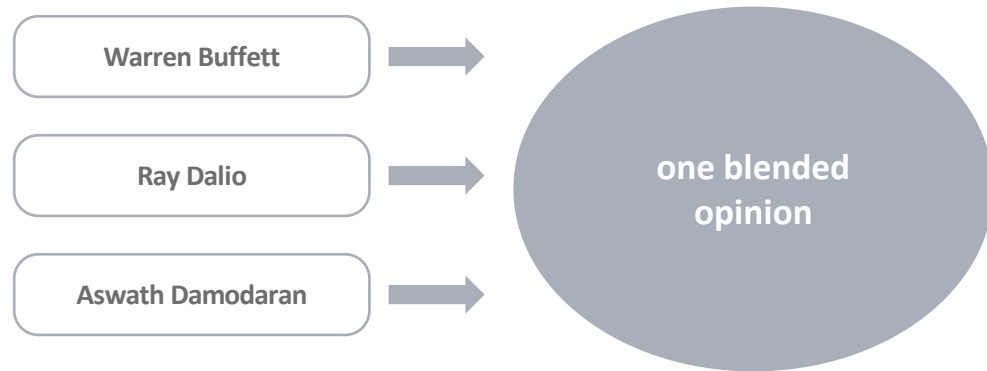
Three layers, two ways to build an analyst, and one surprise about what actually matters.

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Macro / FICC analyst pool · July 2026

When an AI fund loses money, nobody can say why

THE OBVIOUS WAY TO BUILD IT



Each agent is a whole investor with a whole opinion. A weight blends them into one book.

THE MONTH IT LOSES 4%

Which part was wrong?

The belief was inflation really rising?

The trade or was it the wrong way to bet on it?

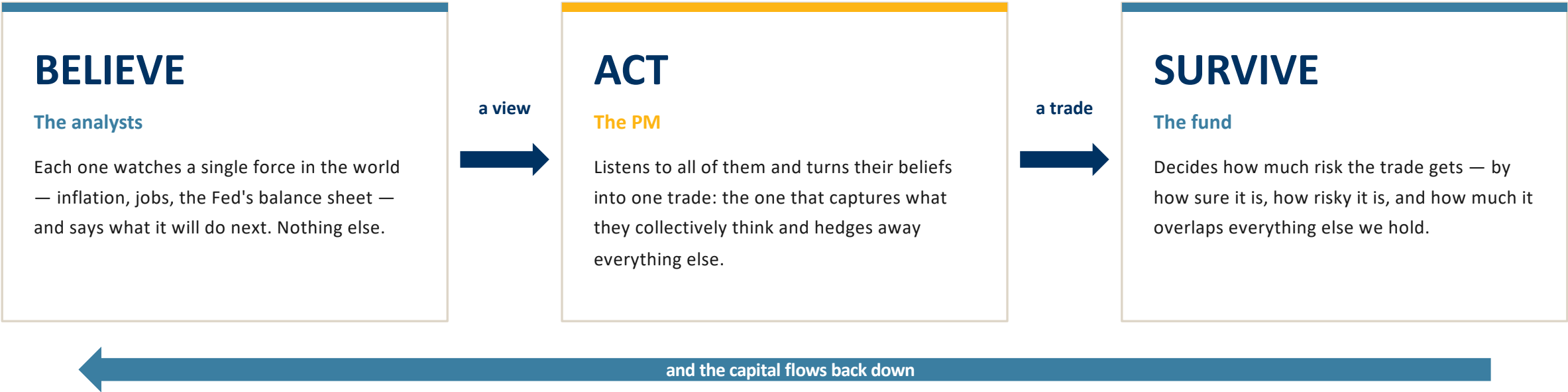
The size or was the bet simply too big?

You cannot know. Nothing in the machine ever recorded the difference.

The opinions arrive pre-mixed. There is no seam to cut along — so there is no question you can ask, and nothing to fix on purpose.

So we gave the fund an org chart

A real fund doesn't blend opinions. It holds a meeting — where different people have different jobs, and each one is answerable for their own.



Today the analysts are macro and rates inflation · jobs · balance sheet · term premium · curve · breakevens · financial conditions equity analysts — teammates, same slot

Each layer talks to the next through one fixed handoff, and never reaches inside the others. **That is the whole discipline.**

An analyst may say “inflation is rising.” It may not say “buy the 10-year.”

WHAT AN ANALYST IS ALLOWED TO SAY

“Inflation is rising.

I'm 46% sure.

It should hold for about a quarter.”

A claim about the world. It can be checked later against what the world did.

WHAT IT MAY NEVER SAY

“Buy the 10-year.”

“Put on a flattener.”

“Go short duration.”

It doesn't know what a flattener is. It has never seen a price. That is on purpose.

only the PM crosses this line

This is what buys us the answer to slide 2. Because the analyst only ever makes a claim about the world, we can grade it on the world — did inflation actually rise? — completely separately from whether the trade made money. When the fund is down, we can ask which layer failed, and get an answer.

Meet one analyst

One driver in, one claim out. Every analyst in the pool is this shape.

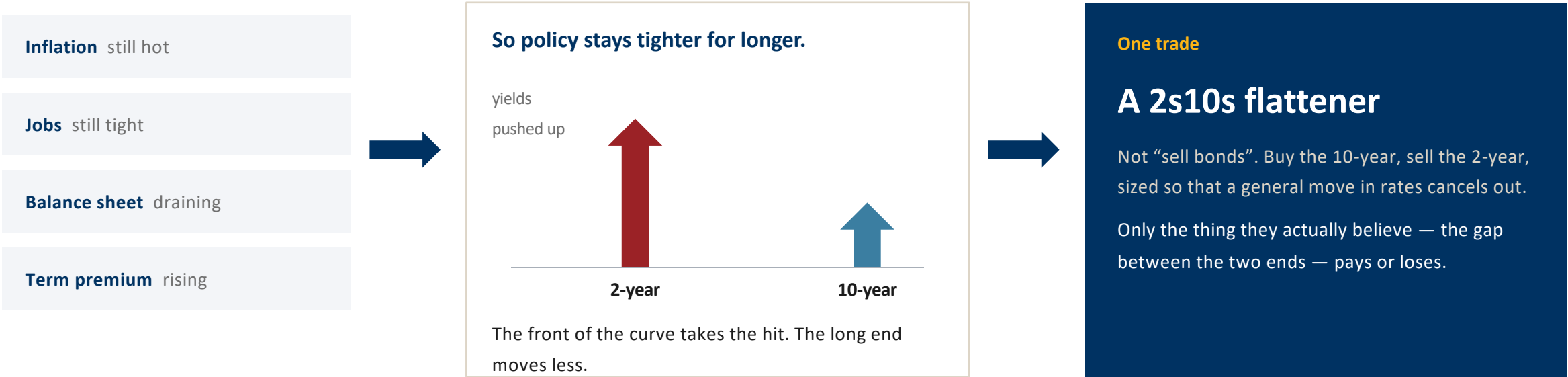


The two inputs do not arrive the same way. The number comes through the gate, and we check on every single run that it obeyed. The Fed's words come in by their own publication date — we test that it never gets tomorrow's statement, but that test runs offline, not on the live run. Worth knowing before slide 11.

Seven analysts, one force each. None of them knows what a trade is.

The PM knows one thing the analysts don't

Four analysts say four things about the world. None of them names a trade. The PM's job — and its whole edge — is knowing how those forces reach the market.



This map is the PM's edge, and it is the part the analysts cannot hold — none of them can see more than one force. **And when they disagree with each other, the PM bets smaller.**

We built every analyst twice

The interesting question isn't whether an AI fund can be built. It's whether the AI is doing any of the work. So each analyst exists in two versions.

THE FORMULA

no AI at all

Read CPI. Measure how much it moved. Say “up”, “down” or “flat”, and how sure you are. About ten lines of arithmetic.

It cannot know the future — no memory, no training data, nothing to recall. Whatever it gets right, it got right honestly.

free • instant

THE MODEL

same job, given to Claude

Hand it the exact same reading and ask for a better view. Same mandate, same output, graded exactly the same way.

It has read the whole internet. On paper, the more sophisticated analyst by a distance.

\$0.94 • 1,252 calls • 47 min

Then we asked the only question that matters: **does the model earn its keep?**

How we grade either of them: we ask whether the driver actually moved the way the analyst said it would, over the next quarter — and we compare that to flipping a coin. 313 weekly meetings, real history, 2019 to 2024.

The model added nothing



HOW MUCH BETTER THAN A COIN FLIP, ACROSS 313 MEETINGS



Given the same numbers to look at, the model just restated what the formula already said.

It didn't do worse. It simply didn't do better — and it disagreed with the formula on only 84 calls out of 1,252.

This is the result that set the direction. If the expensive part of the analyst isn't where the edge comes from, then the lever is somewhere else — and the only other thing an analyst has is what it reads.

So we changed what it reads, not what it is

A CPI print is the past. The Fed says what it is going to do — in words, out loud — before it ever shows up in the data. So we gave the same model three different things to read, and changed nothing else.

NUMBERS

what it read before

The measurement: prices are up 3.1% on the year, and accelerating.

\$0.94

WORDS

the Fed's own statement

The latest FOMC statement, and nothing else. No numbers at all.

\$1.57

BOTH

the number and the story

The measurement and the statement, together.

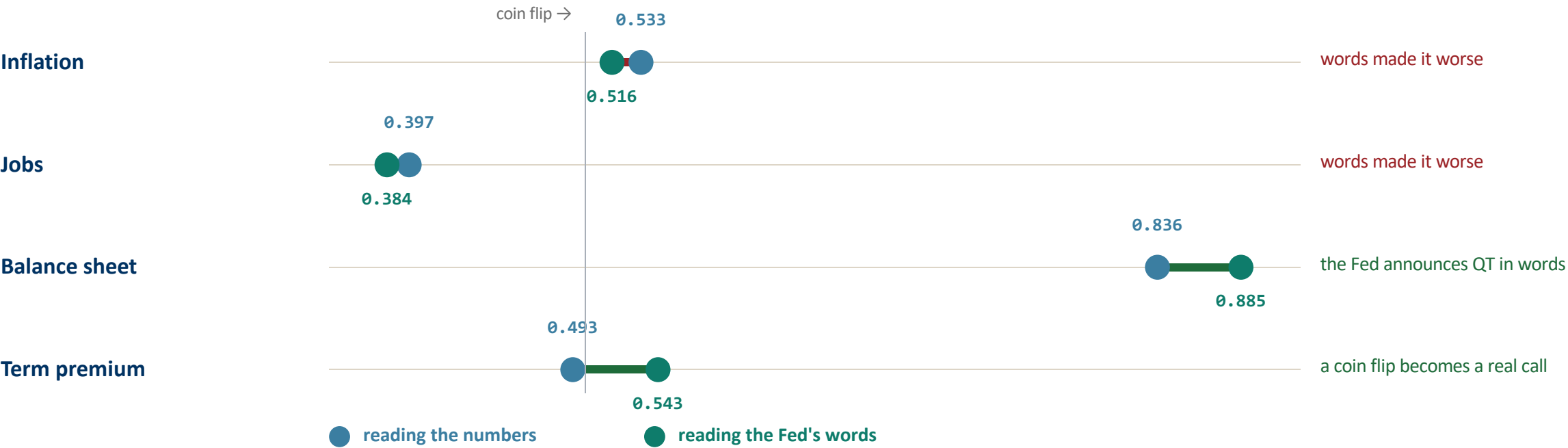
\$1.75

One knob, and we tested that it really is one knob. Same instructions, same analysts, same 313 meetings, same scoring — the three runs differ in nothing but the thing the model reads. We have an offline test that checks each one gets exactly what it should and nothing more, including that it is never handed a statement published after the meeting.

Same brain in all three. So anything that moves is the input, and only the input.

Words help — exactly where the Fed does the talking

HOW OFTEN IT CALLED THE DIRECTION RIGHT · same model, only the reading changes



The pattern isn't “text is better.” It’s “**text is better where text is the source.**”

The Fed announces what it will do to its balance sheet and to long rates. It does not announce next month's CPI.

Being honest about the evidence: all three runs saw the same meetings, so we can check whether a gap is bigger than luck. Only the balance sheet clears that bar — the rest point the right way on 313 meetings, but are not yet proof.

But the words come with a bill

Two things went wrong, and both are worse than a lower hit rate.

Four specialists became one pundit

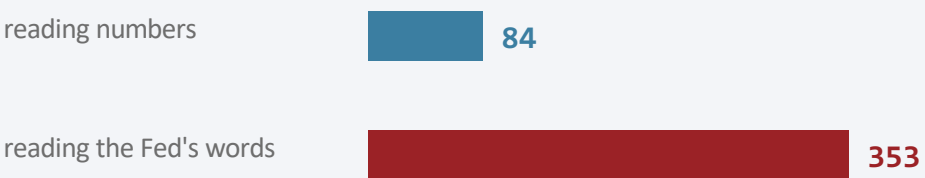
Feed all four analysts the same statement and they all start saying the same thing — because they are all reading the same thing.



The inflation analyst stops tracking inflation. Independence was the whole point of separating them — and the words quietly dissolve it.

And it may be remembering, not reading

The formula cannot know the future. So we count how often the model overrules it — and whether overruling it turns out to be right.



4.2× as many overrules — and it gains most on exactly the two drivers where it “won”. FOMC statements are famous, dated documents. It may simply recall how yields moved next.

We cannot rule this out yet, and we say so: our test window ends before the model's training does, so there is no clean stretch of history it could not have seen. This is a smoke alarm, not a verdict.

Give it both: the number anchors, the words inform

	Better than a coin flip	Still four separate minds?	Suspicious hindsight?
the formula	+0.065	0.238 yes	none — it can't
numbers	+0.065	0.221 yes	84 overrules
words	+0.082	0.339 NO	353 overrules
both	+0.086	0.246 yes	117 overrules

Both together is the only row that wins everywhere.

The measurement pulls each analyst back to its own driver — they are as separate as they were before the AI showed up — and the overrules fall back to a third of what the words alone provoked. The number anchors it. The story informs it.

The input mattered more than the model.

The same Claude sat behind all three columns. Everything that moved, moved because we changed what it was allowed to read.

The whole experiment — three runs, 313 meetings each, 3,756 calls to Claude — cost \$4.25. Everything we have measured so far cost \$7.31.

What we have not tested

We have measured research, not money.

There is no backtest. None.

No P&L, no Sharpe, on any data. The trade construction and the P&L path are written and tested — they have never been run on real history. Everything in this deck grades the analysts, not the fund.

The hindsight alarm is unresolved.

Our window ends in 2024 and the model's knowledge does not. Until we score it on a stretch of history it cannot have read, “it might be remembering” stays on the table.

Three of our seven analysts add nothing.

The curve, breakevens and financial-conditions analysts went in later. One is a coin flip the model overruled 3 times in 313. One is the worst analyst in the pool and loses to “assume last week repeats”.

The words experiment only covers four drivers.

We ran all seven analysts once, on “both” only — so there is no numbers-vs-words comparison for the three newest. And we have never tried the Fed's minutes, only its statements.

None of this is a reason not to show you the results — **it is the reason the results are worth anything.**

What we do next

Equity analysts, from the team

Same slot, same boundary: watch one thing, say what it will do, never name a trade. If the design is right, that costs no change to the PM or the fund.

A second strategy

With one PM there is nothing to diversify and nothing to net. The whole bottom layer is written and idle until there are two books to weigh against each other.

Fix the analysts that don't work

Labor is worse than a coin flip in every version we tried — the input isn't the problem there, the measurement is. Same for breakevens.

Then, finally, a backtest

Once we trust the beliefs, run the book on real history and find out whether good research survives contact with a trade.

The one thing to take away

The layers let us ask which part is wrong. When we asked, the answer wasn't the model — it was what it reads.

Appendix follows: the contracts, the transmission map, and every number behind the last six slides.

The three contracts

The layers talk only through these. Any layer's method can be swapped without touching the others.

DriverView

ANALYST → PM

driver the one force: inflation, term_premium ...

asof the moment the view was formed

direction up / down / flat — about the DRIVER

conviction 0.0 → 1.0

horizon_days how long it should hold

level current reading — makes it scoreable

reasoning why

StrategyTrade

PM → FUND

strategy e.g. macro_rates

legs instrument → signed weight

conviction the PM's confidence in the TRADE

rationale the transmission that produced it

risk what it isolates, what it hedges, net duration

FundAllocation

FUND → PM

capital strategy → multiplier (≥ 0)

constraints strategy → limits

diagnostics netting, vol, breadth

— **no views** — a control layer, not a forecasting one

Inside an analyst, and inside the PM's map

THE INFLATION ANALYST, IN FULL

```
inputs = ("CPIAUCSL",) · horizon_days = 63
Phase 1 – read(world) → DriverView (deterministic, offline)
yoy = cpi.pct_change(12); mom = yoy[-1] - yoy[-4]
direction: up / down / flat, deadband ε = 5bp
conviction = 0.6·|mom|/1% + 0.4·|yoy-2%|/2%
Phase 2 – _refine(view) → DriverView (optional LLM)
One call. System prompt = the persona YAML. User prompt = the reading, the FOMC statement, or both, per --input-mode. Returns JSON {direction, conviction, reasoning}. Any failure returns the Phase-1 view verbatim.
Isolation is structural: the base class hands the analyst an AsOf gate and no other route to data. The text channel is point-in-time by each document's own release_date, not through the gate.
```

THE TRANSMISSION MAP · personas/macro_rates_pm.yaml

	front end (2y)	long end (10y)
inflation	+0.50	—
labor_tightness	+0.35	—
balance_sheet	-0.15	-0.40
term_premium	—	+0.60

slope_pressure = front - long > 0 ⇒ flattener
Legs, DV01-neutral: +IEF (dur 7.5) · -SHY × (7.5/1.9) (dur 1.9), normalized to unit gross ⇒ net_duration ≈ 0.
conviction *= (1 - 0.5 · disagreement) – a split committee bets smaller.
Unit-tested: |net_duration| < 0.15; hawkish ⇒ flattener, dovish ⇒ steepener; book gross ≤ leverage cap.

Every number: accuracy and independence

EDGE OVER A COIN FLIP (hit_rate – 0.5), 313 meetings, 63d horizon, FRED 2019–2024

Driver	n	deterministic	vector	text	text + vector
inflation	304	+0.062	+0.033	+0.016	+0.069
labor_tightness	242	–0.099	–0.103	–0.116	–0.074
balance_sheet	304	+0.312	+0.336	+0.385	+0.349
term_premium	300	–0.017	–0.007	+0.043	+0.000
average edge		+0.0645	+0.0648	+0.0820	+0.0860

own_corr — DOES EACH ANALYST'S VIEW STILL TRACK ITS OWN DRIVER?

Driver	deterministic	vector	text	text + vector
inflation	1.000	0.965	0.469	0.892
labor_tightness	1.000	0.902	0.662	0.939
balance_sheet	1.000	0.881	0.829	0.791
term_premium	1.000	0.962	0.350	0.927
avg cross-correlation ↓ lower = independent	0.238	0.221	0.339	0.246

Faithfulness under text goes negative: inflation –0.171 and term_premium –0.050 — they track other drivers more than their own. Lexicon contamination under text: term_premium 0.850, inflation 0.214. Source: reports/phase1_{vector,text,textvec}.md.

Every number: hindsight, significance, cost

OVERRIDES OF THE DETERMINISTIC CALL · ovr = disagreements on gradeable dates; gain = hit-rate difference

Driver	vector ovr · gain	text ovr · gain	text+vector ovr · gain
inflation	26 · -0.029	107 · -0.046	39 · +0.007
labor_tightness	21 · -0.004	87 · -0.017	13 · +0.025
balance_sheet	29 · +0.024	43 · +0.073	51 · +0.037
term_premium	8 · +0.010	116 · +0.060	14 · +0.017
total	84	353	117

IS THE GAP BIGGER THAN LUCK? · paired McNemar, $z = \text{gain} \times n / \sqrt{\text{Vovr}}$

vs deterministic	gain	ovr	z	
text × balance_sheet	+0.073	43	3.4	significant ($p \approx 0.0007$)
text × term_premium	+0.060	116	1.7	directional
text+vec × labor	+0.025	13	1.7	directional
text+vec × balance_sheet	+0.037	51	1.6	directional
every other cell	—	—	< 1.4	noise

WHAT IT COST — AUDITED

Run	drivers	calls	cost
deterministic	4	0	\$0.00
vector	4	1,252	\$0.94
text	4	1,252	\$1.57
text + vector	4	1,252	\$1.75
text + vector	7	2,191	\$3.05
total		5,947	\$7.31

claude-haiku-4-5 · 0 retries · 17.5h wall clock · 4.51M input tokens

The leak verdict the code computes

Thresholds: a gain ≤ 0.02 is no signal; override_hit needs ≥ 10 overrides to be evidence. Across 12 driver × arm cells: 1 possible leak (text × balance_sheet), 3 small-but-unconfirmed, 8 no signal.

Two caveats

The four older reports still print “possible training-cutoff leak” on every row — a fixed template that never read the numbers, fixed in 7b7b68c but not regenerated. And the 2019–2024 window has no post-cutoff control, so this test is inconclusive by construction.

The seven-analyst run, and what runs offline

ALL SEVEN ANALYSTS · text+vector only, 2,191 calls, \$3.05 — so this is NOT an A/B

Driver	own_corr	hit	edge	gain	ovr
inflation	0.901	0.559	+0.059	-0.003	35
labor_tightness	0.935	0.417	-0.083	+0.016	13
balance_sheet	0.793	0.845	+0.345	+0.033	52
term_premium	0.930	0.500	+0.000	+0.017	14
curve_slope ← new	0.973	0.503	+0.003	+0.003	3
inflation_expectations ← new	0.960	0.321	-0.179	0.000	0
financial_conditions ← new	0.865	0.404	-0.096	+0.089	47

None of the three adds signal.

curve_slope is a coin flip the model overrode 3 times in 313. inflation_expectations has the worst edge in the pool, zero overrides, and loses to persistence by 0.147. financial_conditions has the largest gain (+0.089) but a negative edge — it improves on a bad baseline rather than predicting.

The independence gate bit again. inflation_expectations (T10YIE) correlates 0.558 with term_premium (DGS10) — they share the real rate — and dragged term_premium's faithfulness from 0.698 to 0.442. Same lesson as the real-rates analyst (DFII10) we dropped at 0.86.

Caveat: the lexicon contamination of 1.000 on the three new drivers is an artifact — their mandates name other drivers, and curve_slope's own vocabulary overlaps the trade-term list. The correlation matrix shows no matching coupling.

57 OFFLINE CHECKS · no keys, no network, \$0

test_layered.py 20 checks

AsOf never serves data past the meeting date · each analyst reads only its own series · |net_duration| < 0.15 · hawkish ⇒ flattener, dovish ⇒ steepener · book gross ≤ leverage cap · the scorer returns 1.0 for an always-right analyst, 0.0 for an always-wrong one

test_input_modes.py 9 checks

the A/B is clean: system prompt identical across arms · the vector arm is unchanged whether or not a text source is attached · vector carries no text, text drops the numbers · point-in-time · a document dated asof+5d is never served

test_diagnostics.py 20 checks

the diagnostics harness itself, including the LLM path via a stub that manufactures overrides offline

test_agents.py 8 checks

the legacy flat ensemble degrades safely with no LLM